[INCH-POUND]
A-A-59482
21 December 1999
SUPERSEDING
MIL-L-18937D
13 May 1976

COMMERCIAL ITEM DESCRIPTION

LANTERN, ELECTRIC, HAND, SAFETY

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-L-18937D for all federal agencies.

1. SCOPE. This commercial item description (CID) covers hand-portable, battery-powered safety lanterns to be used for illumination in potentially explosive environments.

2. SALIENT CHARACTERISTICS

2.1 Performance. The lantern shall conform to the performance requirements of the American National Standards Institute (ANSI/UL) Standard 783, "Electric Flashlights and Lanterns for Use in Hazardous (Classified) Locations".

2.1.1 Environment. The lantern shall operate safely in hazardous environments as defined by the National Fire Protection Association (NFPA) Standard 70, "National Electrical Code Handbook", for Class I, Division I, Groups C and D (-1) or Group D (-2) only (see 6.1).

2.1.2 Center beam. The specified lamp and reflector shall project a beam of light with its maximum luminous intensity at the approximate center of the beam. When tested in a totally darkened environment with a fully charged battery, the center beam of maximum intensity shall be 10 to 20 inches in diameter at a distance of 10 feet from the lantern lens.

2.1.3 Luminous intensity. Beam candlepower shall be determined by averaging the peak reading of the center beam with four peripheral readings at 90-degree intervals taken at the approximate midpoint between the center beam radius and the outer edge of the full beam. The average beam illumination measured by a calibrated photometer at a distance of 10 feet with the beam axis perpendicular to the photometer sensor shall be at least 300 foot-candles. The beam luminous

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AMSC N/A FSC 6230

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intensity (I) in candlepower shall be calculated using the formula $I = ED^2$, where E is the average photometer reading in foot-candles and D is the distance in feet from the photometer to the lantern lens.

2.1.4 Beam direction. Light beam direction shall be vertically adjustable through an arc ranging from -10 degrees to +90 degrees, relative to the lantern case.

2.1.5 Power consumption. Replenishment with a new battery shall sustain continuous lantern operation at rated beam candlepower for at least four hours.

2.2 Components. Lantern components shall include a case, lamp housing, switch, carrying handle, and any other necessary parts. The assembled lantern weight (including battery) shall not exceed 10 pounds.

2.2.1 Case. The lantern case and all exposed parts shall be non-sparking, non-magnetic, and able to withstand heat up to 230 ^{o}F without deformation. Plastic parts shall also withstand heat up to 230 ^{o}F in accordance with the American Society for Testing and Materials (ASTM) Standard D648, "Deflection Temperature Under Flexural Load". Contact between dissimilar metals shall be avoided to minimize galvanic corrosion. The case structure shall support 20 pounds of weight without deformation and the lantern shall remain capable of producing rated illumination after being subjected to one drop on each of its six planes from a height of five feet onto a steel or concrete surface. Bulb breakage or battery damage resulting from any drop will require a repeat of the drop but shall not be cause for rejection.

2.2.2 Batteries. The lantern shall operate with standard commercially available dry-cell batteries conforming to the American National Standards Institute (ANSI) Standard C18.1, "Specifications for Dry Cells and Batteries".

2.2.3 Lens. A shatter-resistant lens shall protect the lamp socket and reflector from dirt and moisture.

2.2.4 Lamp. At least one spare lamp shell be shock-mounted inside the lantern case. Standard sized replacement lamps shall be commercially available.

2.2.5 Switch. The switch shall be spring-loaded for positive engagement, shall be sealed against contamination, and shall be easily actuated by operators wearing heavy gloves. Either toggle or pushbutton type switches are acceptable.

2.2.6 Markings. The lantern shall be marked in accordance with ANSI/UL 783 (see 6.3(b)).

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3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials, to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The lanterns provided shall meet the salient characteristics of this commercial item description; shall conform to the producer's own drawings, specifications, standards, and quality assurance practices; and shall be the same product as that which is offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The lanterns offered must have previously been sold to the commercial market or government.

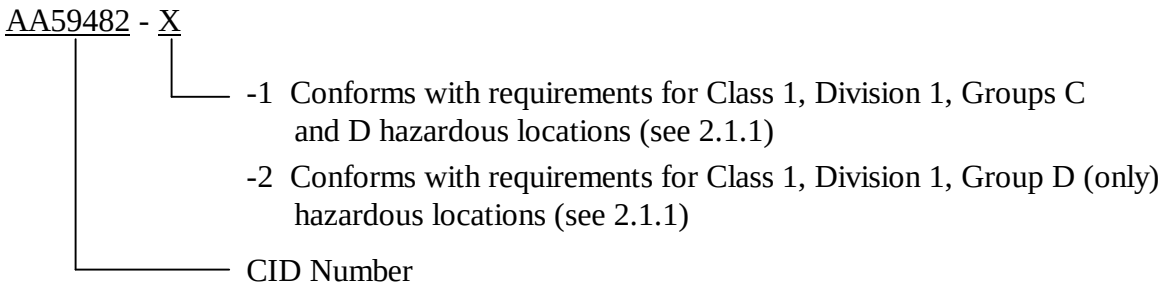
4.3 Warranty. Unless otherwise specified in the contract, the manufacturer's standard commercial warranty terms shall apply.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order (see 6.3(c)).

6. NOTES

6.1 Part identification number (PIN). The part identification numbering system shown below is intended for government purposes and does not constitute a requirement for the contractor.



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6.2 Sources of documents.

6.2.1 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 W 42nd Street, New York, NY 10036.

6.2.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

6.2.3 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20401-0001.

6.2.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.

6.3 Ordering data. Acquisition documents must specify:

a. Title, number, and date of this document
b. Marking requirements, if different (see 2.2.6)
c. Packaging requirements (5.1)

6.4 National stock number (NSN). The NSN 6230-00-006-1897 (CAGE Code 61222) is assigned for the lantern specified in this CID. Other NSNs may also correspond with this CID.

MILITARY INTERESTS:

Custodians:
Air Force - 99
Navy - CG

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS

(Project 6230-1084)

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